

INDUSTRY & ACADEMIC EXPERIENCE

Google Research

Research Engineer

Feb 2022 - *present*

- My research is focused on foundational machine learning theory and its application to improving efficiencies in training and fine-tuning large-scale models as well as model safety and fairness.

Google Cloud

AI Engineer

Apr 2018 - Feb 2022

- Led scoping and delivery for ML/AI engagements with Cloud customers.
- Delivered dedicated ML Instruction for Google's Advanced Solutions Lab

New York University, Courant Institute

Clinical Assistant Professor

Aug 2014 - May 2017

University of Missouri, Department of Mathematics

Assistant Professor

Aug 2011 - May 2014

PUBLICATIONS (BOOKS)

1. **Michael Munn**, David Pitman. *Explainable AI for Practitioners: Designing and Implementing Explainable ML Solutions*. O'Reilly, 2022.
2. Valliappa Lakshmanan, Sara Robinson, **Michael Munn**. *Machine Learning Design Patterns: Solutions to Common Challenges in Data Preparation, Model Building and MLOps*. O'Reilly, 2020.

PUBLICATIONS (ARTICLES)

* = equal contribution

1. Adrian Goldwaser, **Michael Munn**, Javier Gonzalvo, Benoit Dherin. *Equivalence of Context and Parameter Updates in Modern Transformer Blocks*. ICML, 2026 (oral, spotlight).
2. Benoit Dherin*, **Michael Munn***, Javier Gonzalvo, Blaz Bratanić, Ananth Balashankar, Adrian Goldwaser, Mor Geva. *Mechanistic Jailbreaking: Optimizing Universal Defenses via the Safety Operator*. In preparation.
3. Benoit Dherin*, Benny Avelin*, Anders Karlsson*, Hanna Mazzawi, Javier Gonzalvo, **Michael Munn**. How iteration order influences convergence and stability in deep learning. TMLR, 2025.
4. Benoit Dherin*, **Michael Munn***, Hanna Mazzawi*, Michael Wunder, Javier Gonzalvo. *Learning without Training: The implicit dynamics of in-context learning*. In review.
5. Benoit Dherin*, **Michael Munn***. On residual network depth. In review.
6. **Michael Munn***, Susan Wei*. *A Bayesian model selection criterion for selecting pretraining checkpoints*. ICML, 2025.
7. **Michael Munn***, Benoit Dherin*, Javier Gonzalvo. *The Impact of Geometric Complexity on Neural Collapse in Transfer Learning*. NeurIPS, 2024.
8. Benoit Dherin*, **Michael Munn***, Hanna Mazzawi, Michael Wunder, Sourabh Medapati, Javier Gonzalvo. *Learning by solving differential equations*. NeurIPS, OPT Workshop, 2025
9. **Michael Munn***, Benoit Dherin, Javier Gonzalvo. *A margin-based multiclass generalization bound via geometric complexity*. ICML, 2023.

- PUBLICATIONS
(ARTICLES)
CONT'D
1. Ryan Gillard, Stephen Jonany, Yingjie Miao, **Michael Munn**, Connal de Souza, Jonathan Dungay, Chen Liang, David R. So, Quoc V. Le, Esteban Real. *Unified Functional Hashing in Automated Machine Learning*.
 2. Benoit Dherin*, **Michael Munn***, Mihaela Rosca, David G.T. Barrett. *Why neural networks find simple solutions: the many regularizers of geometric complexity*. NeurIPS, 2022.
 10. Benoit Dherin*, **Michael Munn***, David G.T. Barrett. *The Geometric Occam's Razor Implicit in Deep Learning*. NeurIPS, OPT Workshop 2021.
 11. Tianlin Xu, Li Kevin Wenliang, **Michael Munn**, Beatrice Acciaio. *COT-GAN: Generating Sequential Data via Causal Optimal Transport*. NeurIPS, 2020.

- PUBLICATIONS
(PRE-GOOGLE)
- author order is alphabetical*
1. Sajjad Lakzian, **Michael Munn**. *On the Size of a Ricci Flow Neckpinch via Optimal Transport*. Analysis and Geometry of Metric Spaces, 2021.
 2. Qintao Deng, Fernando Galaz-García, Luis Guijarro, **Michael Munn**. *Three-Dimensional Alexandrov spaces with positive or nonnegative Ricci curvature*. Potential Analysis 2017.
 3. Lashi Bandara, Sajjad Lakzian, **Michael Munn**. *Geometric singularities and a flow tangent to the Ricci flow*. Annali S.N.S di Pisa, 2015.
 4. **Michael Munn**. *Alexandrov spaces with large volume growth*. Journal of Mathematical Analysis and Applications, 2015.
 5. Sajjad Lakzian, **Michael Munn**. *Super Ricci flow for disjoint unions*. Analysis and Geometry of Metric Measure Spaces, 2012.
 6. **Michael Munn**. *Volume growth and the topology of pointed Gromov-Hausdorff limits*. Differential Geometry and Its Applications, 2010.
 7. **Michael Munn**. *Volume growth and the topology of manifolds with nonnegative Ricci curvature*. J. of Geom. Analysis, 2010.
 8. Dan Garbin, Jay Jorgenson, **Michael Munn**. *On the appearance of Eisenstein series through degeneration*. Commentarii Mathematici Helvetici, 2008.

EDUCATION

<u>University of Warwick</u>	Coventry, UK
NSF Postdoctoral Fellow	
• Research areas: optimal transport, metric measure spaces, Ricci flow	
<u>City University of New York</u>	New York, NY
<i>Ph.D. in Mathematics</i>	
• Research areas: geometric analysis, topology	
<u>University of Notre Dame</u>	South Bend, IN
<i>B.S. in Honors Mathematics</i>	
• Magna Cum Laude	

VOLUNTEERING / ADVISING	<u>Google.org</u> GenAI Accelerator, AI Coach • developed an agent for antibiotic discovery using Google’s ADK with biotech startup Phare-Bio. • built a Gemini-based lab assistant for development and evaluation of a bio-materials with UK-based NGO Materiom. • advised UK-based FullFact team in building a Gemini tool to monitor and analyze online videos to surface misinformation and harmful health claims <u>Google.org</u> Tech Lead, The Trevor Project • led a team of ML SWEs across Google and Trevor, coordinated with business stakeholders to align on delivery and key objectives • developed a custom conversation simulator model using PyTorch and GPT2/GPT3 to be used by Trevor trainees • trainees used our conversation simulator for 4,919 hrs of training, removing 35-50 role-play shifts for trainers and the potential to graduate an additional 200-300 new counselors per year
SKILLS	<u>Programming:</u> Python, SQL, (beginner) C++. <u>Tools, Skills, Areas of Interest:</u> JAX/Flax, Tensorflow/Keras, PyTorch, Scikit-Learn, Text Generation (GPT, Transformers), Image Generation (GANs, diffusion models), fine-tuning and transfer learning, explainable/interpretable AI and XAI tools (Captum, SHAP, LIME, LIT), statistical learning theory and generalization, implicit regularization, Riemannian geometry, optimal transportation, Google Cloud Platform, AWS, Kubeflow/Kubernetes, Big-Query, Dataflow, Colab/Jupyter, Flask.
ACADEMIC SERVICES	<u>Reviewer for:</u> <i>NeurIPs</i> 2020, 2022, 2023, 2024, 2025 <i>ICML</i> 2023, 2024, 2025 <i>ICLR</i> 2025